

M50: Final Composite Dependent Variable

PCA on Raw Switch Counts (Time, Topic-LDA, Typing)

No standardization (z-score) applied

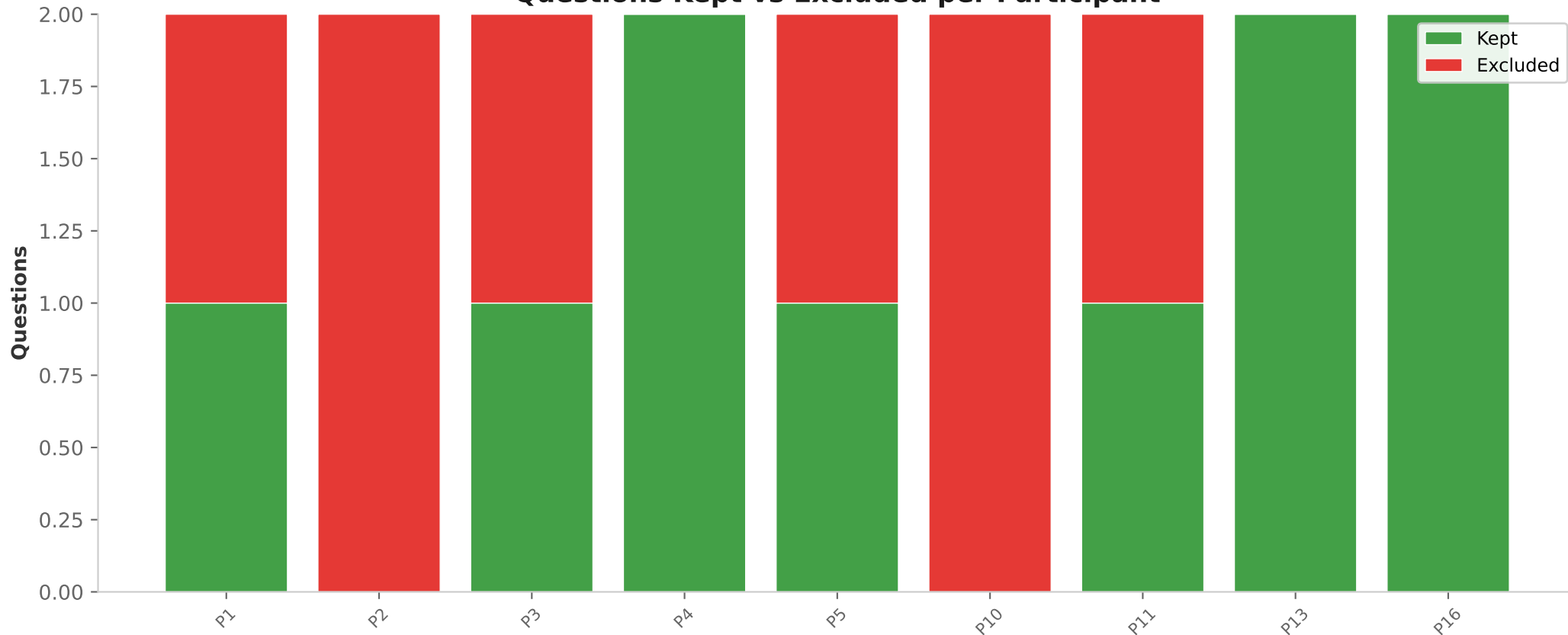
Exclusion Criteria Applied:

1. Questions with fewer than 3 page visits: 5 excluded
2. Questions with idle time $\geq 50\%$ after last meaningful event: 6 excluded
3. Participants > 3 SD from mean on any DV: 0 excluded

Result: 10 / 18 questions retained

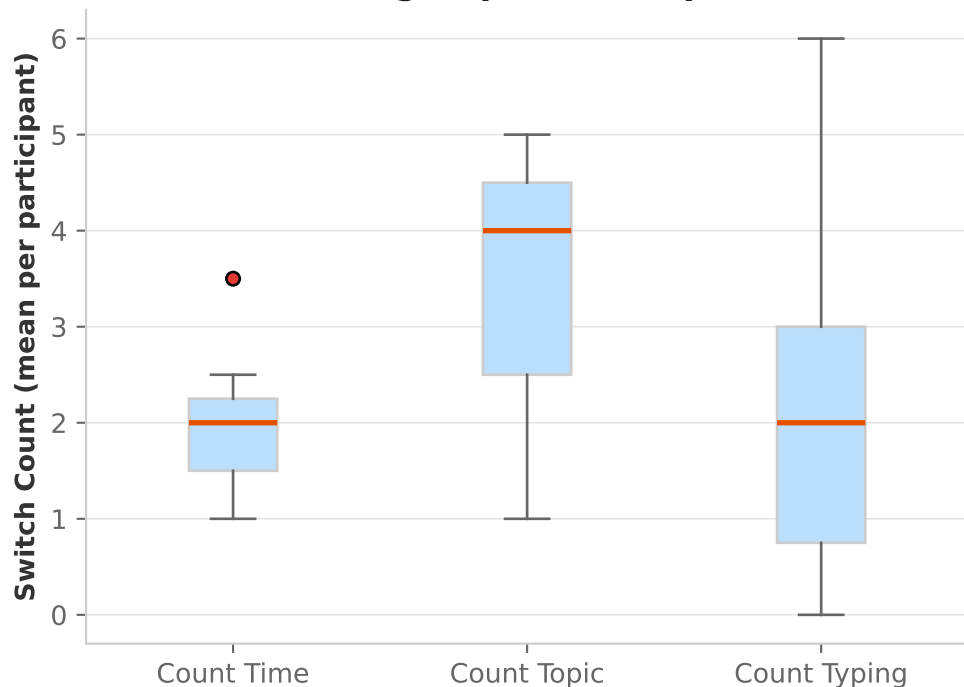
Final sample: N = 7 participants

Exclusion Summary: Questions Kept vs Excluded per Participant

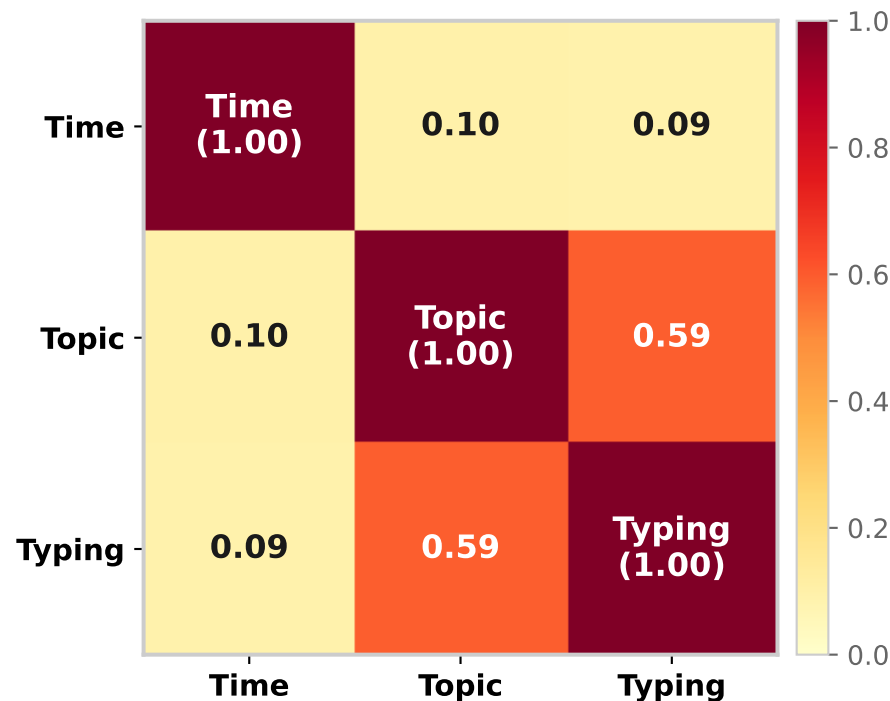


Descriptive Statistics: Raw Switch Counts

Distribution of Raw Switch Counts (Averaged per Participant)

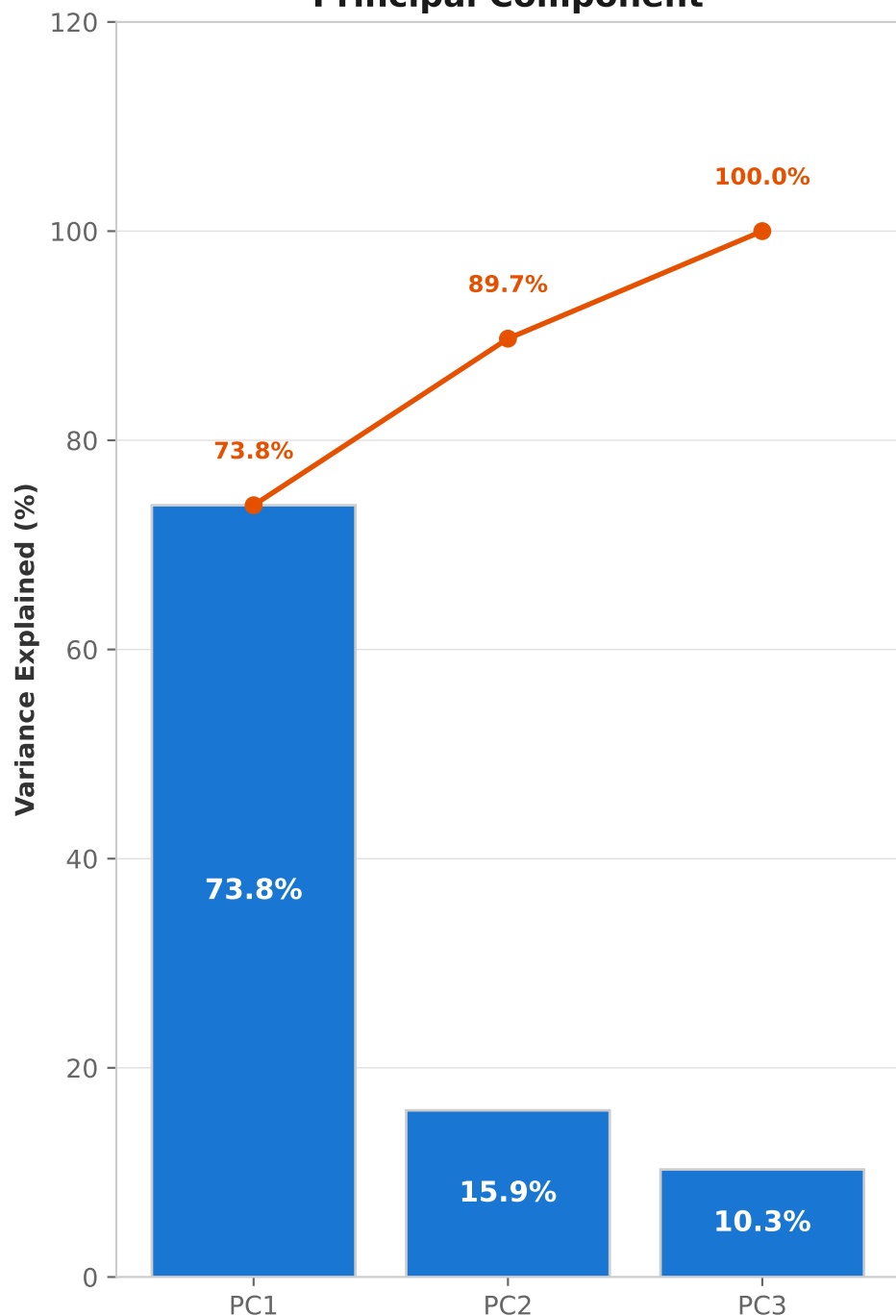


Correlation Between Switch Count Variables



PCA Results: Dimensionality Reduction on Raw Switch Counts

Variance Explained by Each Principal Component



PCA Loadings (variable weights per component):

PC1 (73.8% variance):

Count Typing	+0.862	+++++
Count Topic	+0.505	+++++
Count Time	+0.045	

PC2 (15.9% variance):

Count Topic	+0.860	+++++
Count Typing	-0.506	-----
Count Time	+0.058	+

PC3 (10.3% variance):

Count Time	+0.997	+++++
Count Topic	-0.073	-
Count Typing	-0.010	

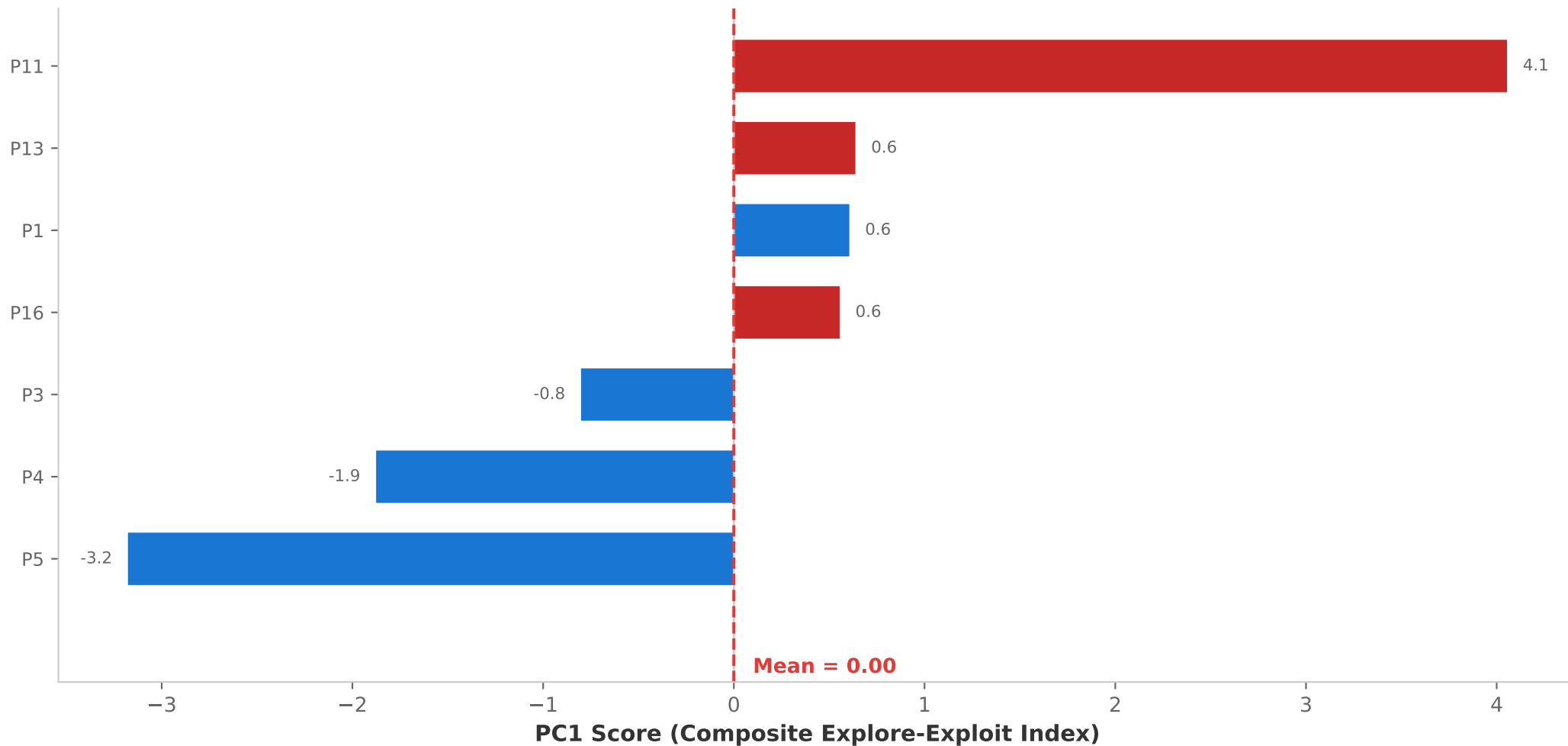
Interpretation:

PC1: Primarily reflects typing switching
(loading = +0.862)
Higher score = more typing transitions

PC2: Primarily reflects topic switching
(loading = +0.860)
Higher score = more topic transitions

PC3: Primarily reflects time switching
(loading = +0.997)
Higher score = more time transitions

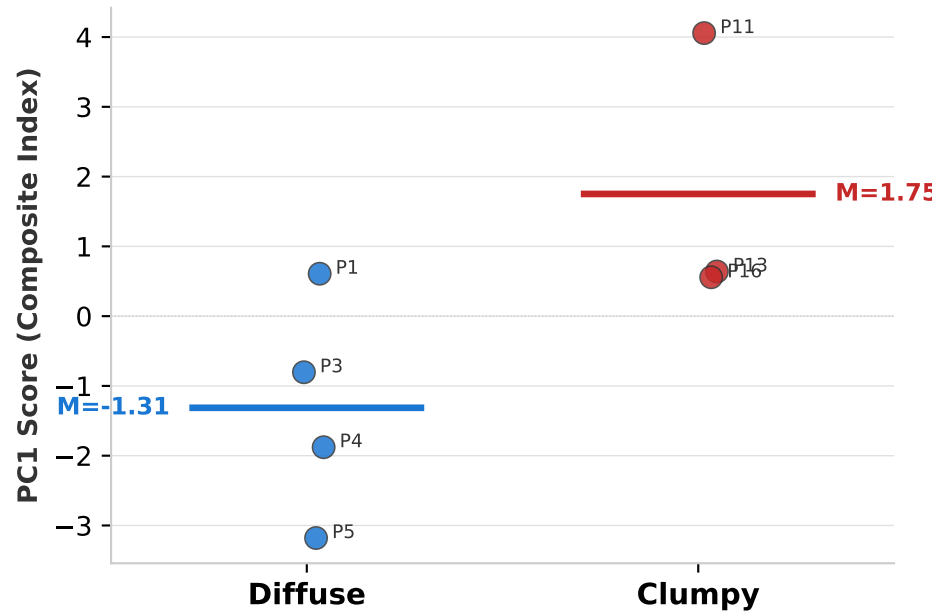
Individual PC1 Scores: Composite Explore-Exploit Index



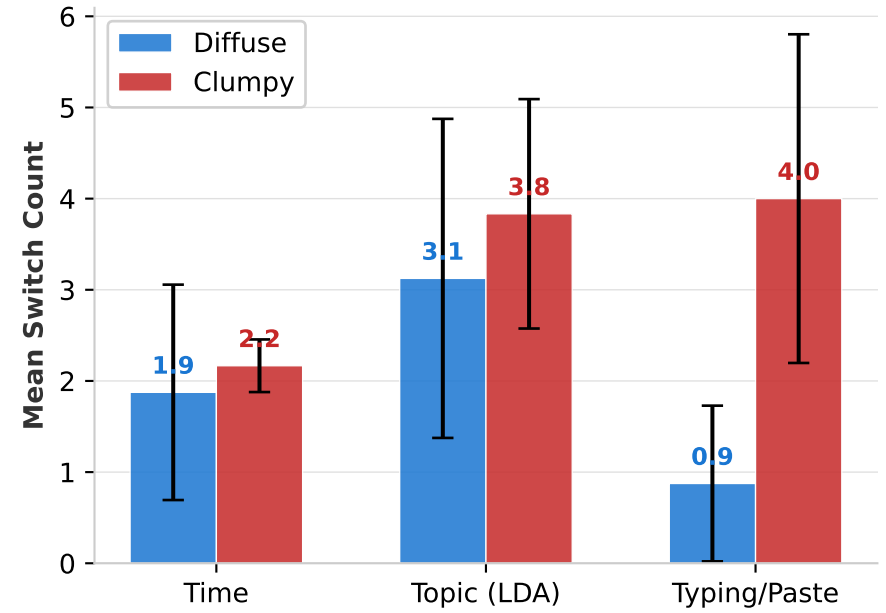
Condition Comparison: Clumpy vs Diffuse

N = 7 (Diffuse: 4, Clumpy: 3)

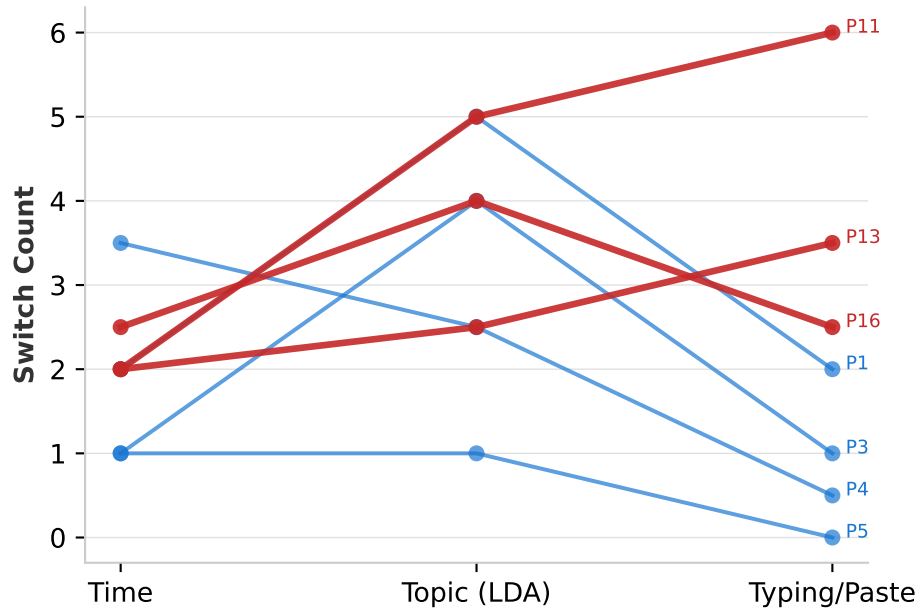
PC1: Composite Explore-Exploit
by Condition



Raw Switch Counts
by Condition (Mean +/- SD)



Individual Profiles
by Switch Type



Conditions

- Diffuse (n=4)
- Clumpy (n=3)

Condition Comparison: Descriptive Statistics & t-Tests

N = 7 (Diffuse: 4, Clumpy: 3)

DESCRIPTIVE STATISTICS

Variable	Diffuse (n=4)	Clumpy (n=3)
Time	M = 1.88, SD = 1.18	M = 2.17, SD = 0.29
Topic (LDA)	M = 3.12, SD = 1.75	M = 3.83, SD = 1.26
Typing/Paste	M = 0.88, SD = 0.85	M = 4.00, SD = 1.80
PC1 (composite)	M = -1.31, SD = 1.61	M = 1.75, SD = 2.00

INDEPENDENT-SAMPLES t-TESTS (Welch)

Variable	t	df	p	Cohen d	Sig.
Time	-0.48	3.5	0.6630	+0.31	n.s.
Topic (LDA)	-0.62	5.0	0.5607	+0.45	n.s.
Typing/Paste	-2.78	2.7	0.0786	+2.37	n.s.
PC1 (composite)	-2.18	3.8	0.0980	+1.73	n.s.

* p < .05 ** p < .01 *** p < .001

INTERPRETATION

PC1 difference (Clumpy - Diffuse): +3.06
Clumpy condition shows more explore-exploit switches
Effect size (Cohen's d): +1.73
Result is NOT statistically significant (p = 0.0980)
Low statistical power due to small sample (N = 7)

Largest sub-variable effect: Typing/Paste
d = +2.37, p = 0.0786

Note: Typing metric relies on paste events only
(answer_snapshot missing from current log).